

21PYB10BJ- Question Bank

Module-I

Part B

1. What are the postulates of classical free electron theory?
2. Write the success of classical free electron theory.
3. Write any three failures of classical free electron theory.
4. What are the postulates of quantum free electron theory?
5. Write the success of quantum free electron theory.
6. Write a short note on splitting energy band in solids.
7. Explain three types of E-K diagram.
8. What is meant by Brillouin Zone? Draw the diagram for Brillouin zone in 1D and 2D Lattice.
9. Write any three differences between direct and indirect band gap semiconductor.
10. Write a short note on Phonons.
11. Explain probability of occupation in a given energy level using Fermi-Dirac distribution.
12. Explain direct band gap and indirect band gap in materials with the help of E-K diagram.
13. Write the classification of electronic materials on the basis of band theory.

Part C

1. Kronig-Penney Model
2. Density of states

Module-II Question Bank

1. What is intrinsic semiconductor?
2. Explain atomic structure and energy level diagram of intrinsic semiconductor?
3. Where the Fermi energy level lies in intrinsic semiconductor. How does it vary with respect to temperature?
4. What is extrinsic semiconductor?
5. Explain N - type semiconductor with diagram
6. Explain P - type semiconductor with diagram
7. Where the Fermi energy level lies in P - type semiconductor. How does it vary with respect to temperature?
8. Where the Fermi energy level lies in N - type semiconductor. How does it varies with respect to temperature.
9. Describe the difference between P-type and N-type semiconductor materials?
10. Explain about photon transitions
11. Explain about SRH carrier generation and recombination.
12. Explain Auger transitions.
13. Explain the process of impact ionization.

- 14.** Discuss drift current and derive the equation for J_d (drift).
- 15.** Discuss about diffusion current and obtain the equation for J_d (diffusion).
- 16.** Derive the continuity equation for an electron with an infinitesimal slice thickness of dx in a one-dimensional system.
- 17.** Explain the concepts of built in potential in p-n junction under thermal equilibrium.
- 18.** Describe the change in potential under Forward bias condition in p-n junction.
- 19.** Describe the change in potential under Reverse bias condition in p-n junction.
- 20.** Discuss the VI characteristics of p-n junction diode under forward and reverse biasing.